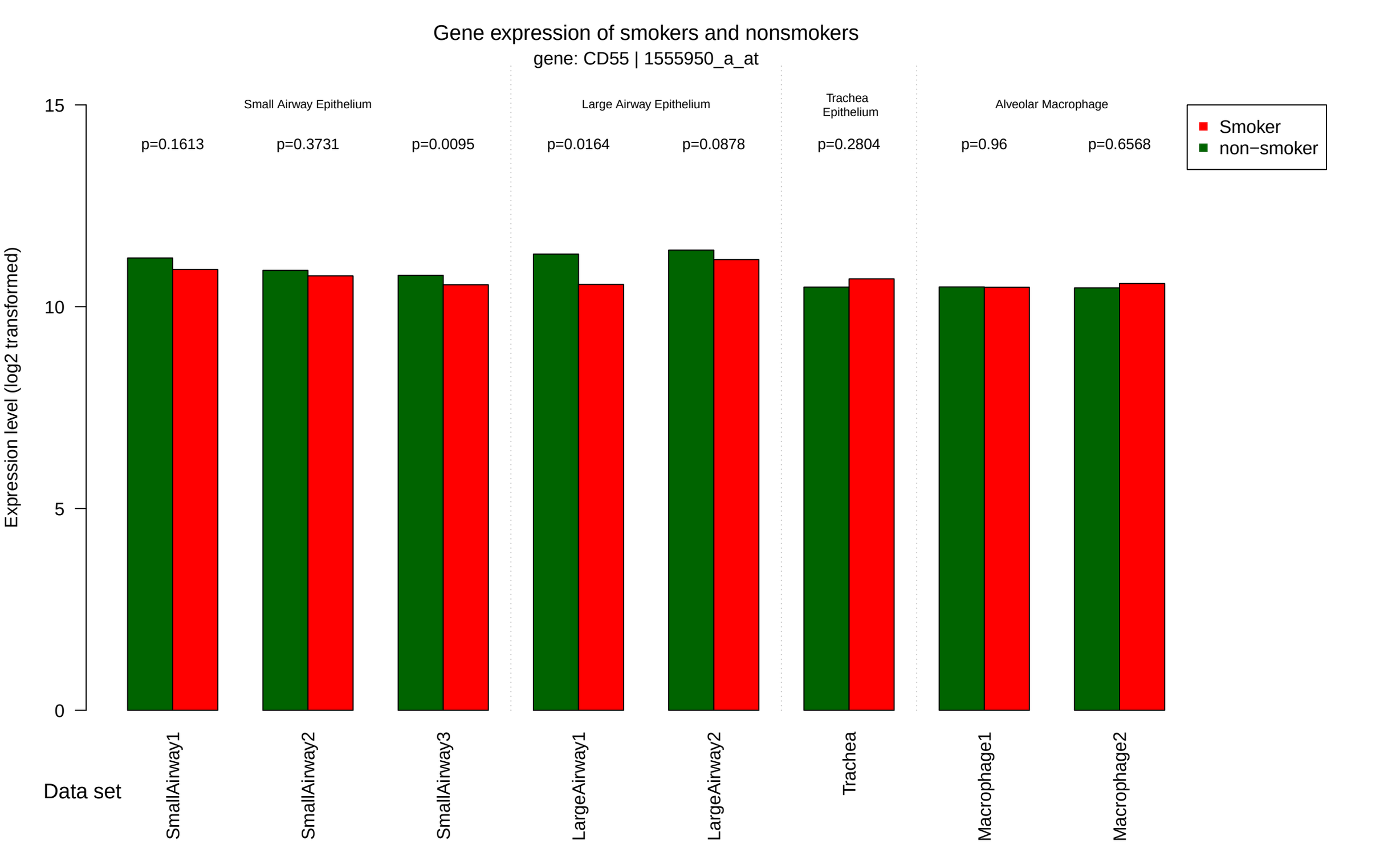
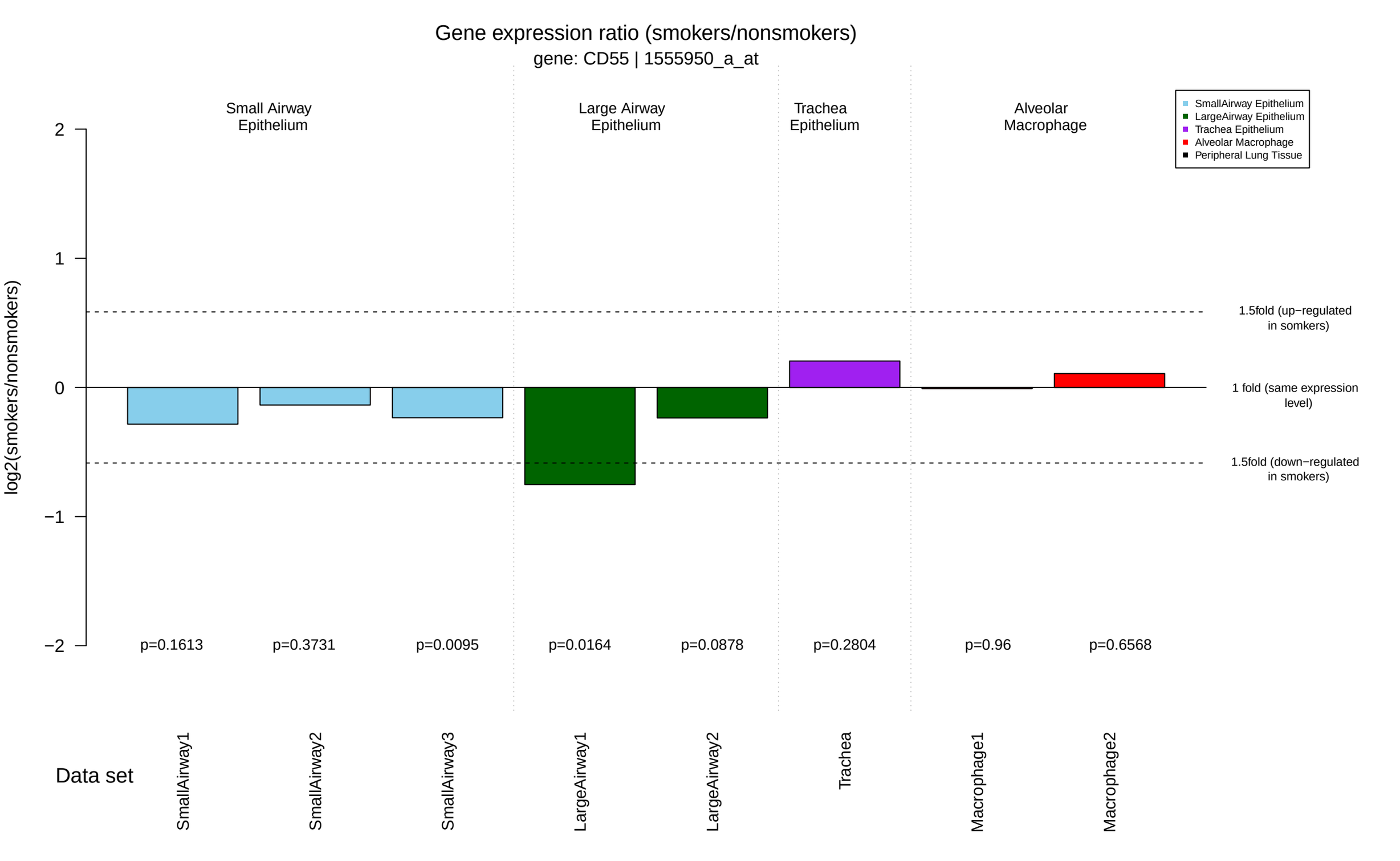
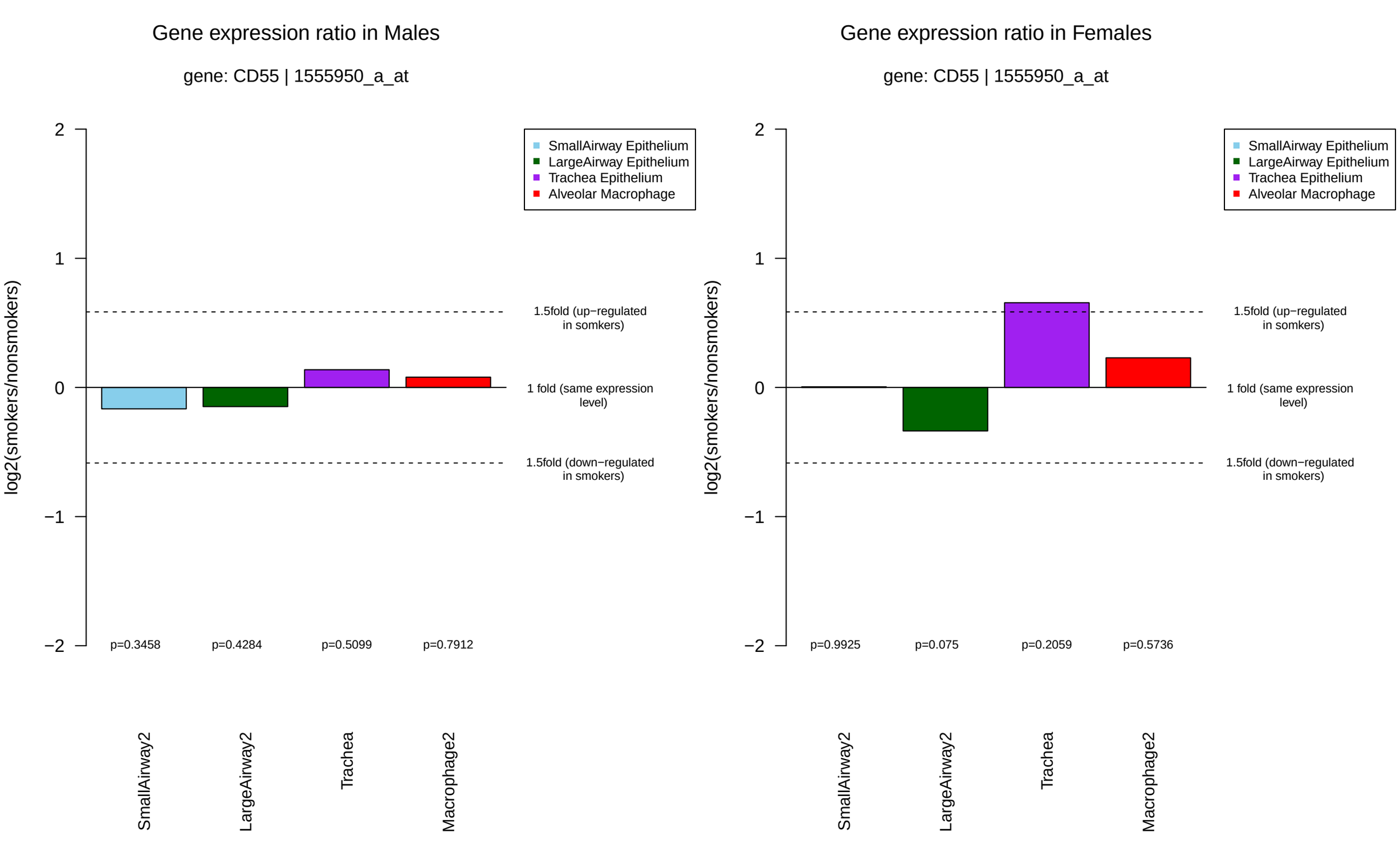
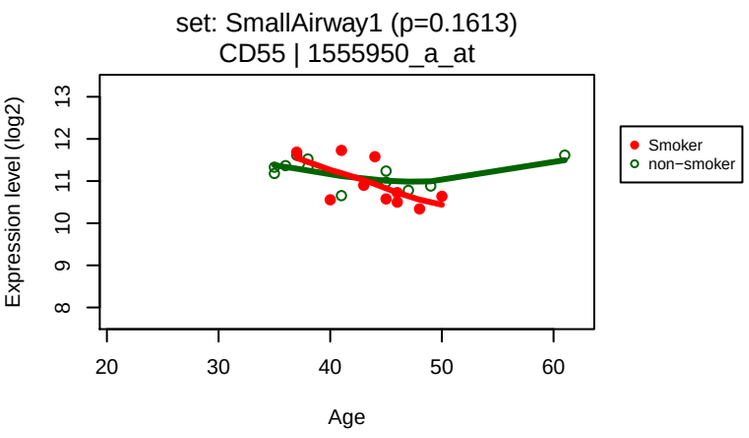
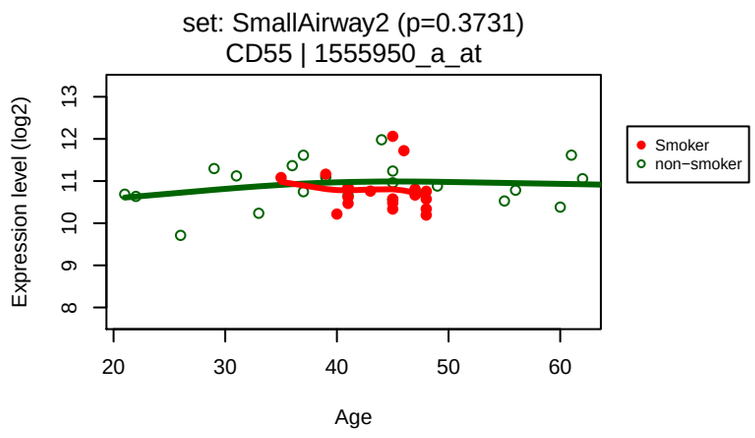
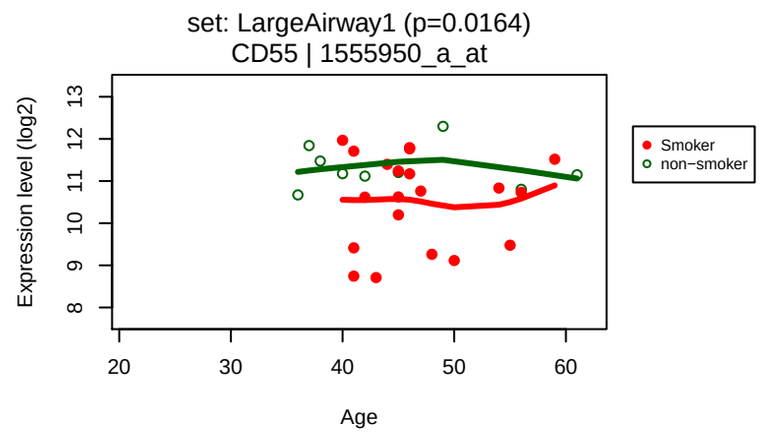

Gene expression of smokers and nonsmokers
gene: CD55 | 1555950_a_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
p=0.1613
p=0.3731
p=0.0095
p=0.0164
p=0.0878
p=0.2804
p=0.96
p=0.6568
Smoker
non−smoker
Expression level (log2 transformed)
15
10
5
0
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

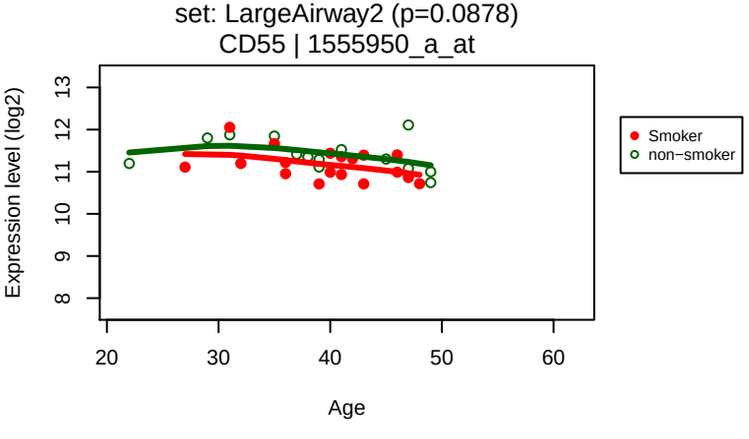

Gene expression ratio (smokers/nonsmokers)
gene: CD55 | 1555950_a_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Peripheral Lung Tissue
log2(smokers/nonsmokers)
2
1
0
−1
−2
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.1613
p=0.3731
p=0.0095
p=0.0164
p=0.0878
p=0.2804
p=0.96
p=0.6568
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

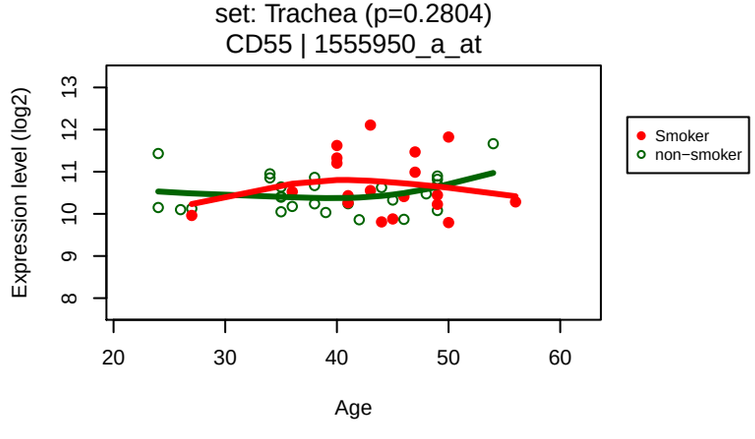

Gene expression ratio in Males
gene: CD55 | 1555950_a_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.3458
p=0.4284
p=0.5099
p=0.7912
SmallAirway2
LargeAirway2
Trachea
Macrophage2
Gene expression ratio in Females
gene: CD55 | 1555950_a_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.9925
p=0.075
p=0.2059
p=0.5736
SmallAirway2
LargeAirway2
Trachea
Macrophage2

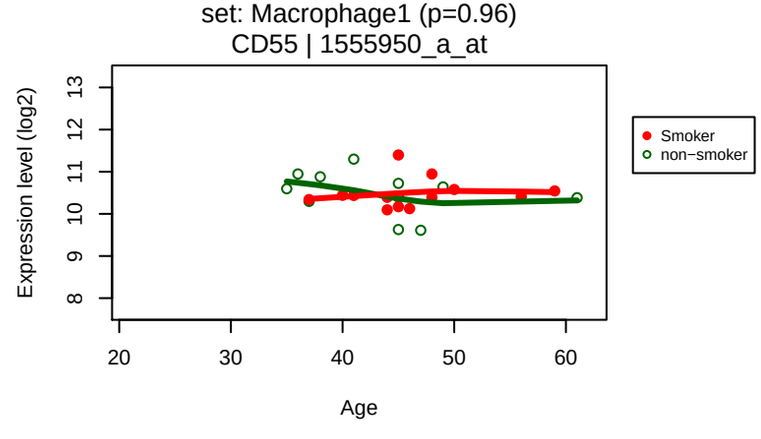

set: SmallAirway1 (p=0.1613)
CD55 | 1555950_a_at
Expression level (log2)
Age
Smoker
non−smoker

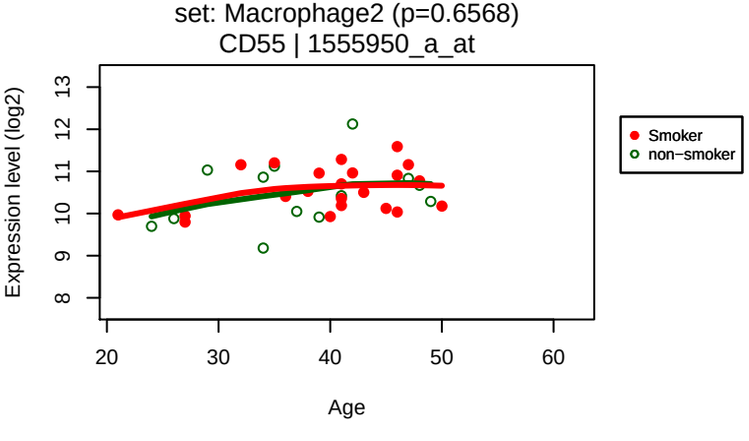

set: SmallAirway2 (p=0.3731)
CD55 | 1555950_a_at
Expression level (log2)
Age
Smoker
non−smoker

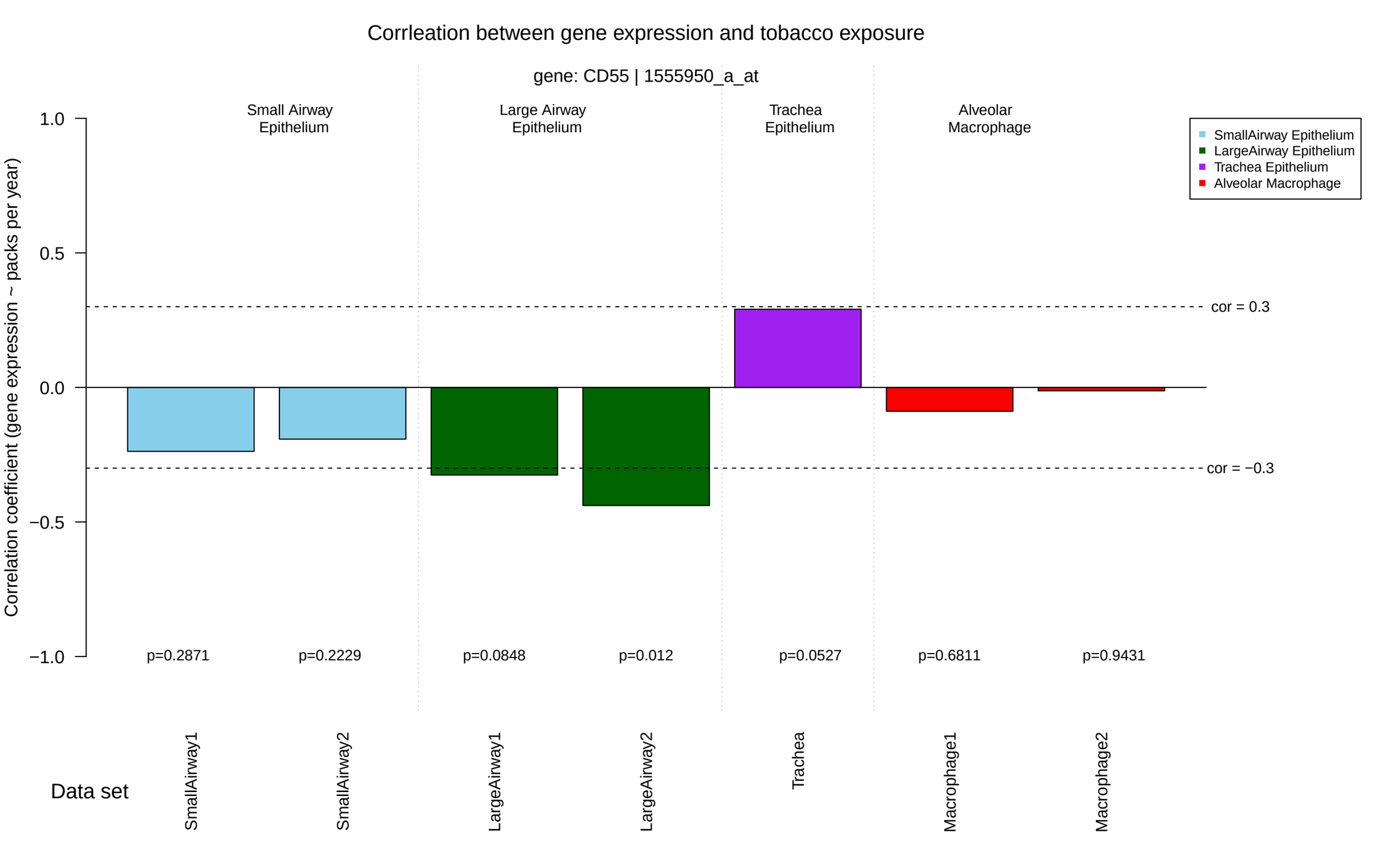

set: LargeAirway1 (p=0.0164)
CD55 | 1555950_a_at
Expression level (log2)
Age
Smoker
non−smoker


set: LargeAirway2 (p=0.0878)
CD55 | 1555950_a_at
Expression level (log2)
Age
Smoker
non−smoker


set: Trachea (p=0.2804)
CD55 | 1555950_a_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage1 (p=0.96)
CD55 | 1555950_a_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage2 (p=0.6568)
CD55 | 1555950_a_at
Expression level (log2)
Age
Smoker
non−smoker


Corrleation between gene expression and tobacco exposure
gene: CD55 | 1555950_a_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Correlation coefficient (gene expression ~ packs per year)
1.0
0.5
0.0
−0.5
−1.0
cor = 0.3
cor = −0.3
p=0.2871
p=0.2229
p=0.0848
p=0.012
p=0.0527
p=0.6811
p=0.9431
Data set
SmallAirway1
SmallAirway2
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2